

CABAUF TOWER, Netherlands

WMO: 063480

Lat: 51.9703N

Lon: 4.9263E

Elev: 7

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.2	(c) 24.6	(d) 15.5	(e) 12.1	(f) 24.4	(g) 20.0	(h) 15.0	(i) 26.1	(j) 33.7	(k) 48.3	(l) 30.1	(m) 48.6	(n) 6.3	(o) 70	(p) 0.578

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 15.2	(c) 82.6	(d) 68.8	(e) 78.7	(f) 67.1	(g) 75.5	(h) 65.2	(i) 70.5	(j) 79.8	(k) 68.5	(l) 76.5	(m) 66.7	(n) 73.4	(o) 7.7	(p) 110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	100.2	74.8	65.5	94.2	72.2	64.0	89.2	70.2	34.4	80.0	32.7	76.8	31.3	73.3	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.7	23.0	20.1	16.4	90.3	6.8	3.2	11.5	92.6	7.5	94.5	3.7	96.3	-1.2	98.6
			15.6	75.2	6.7	2.0	10.8	76.6	6.9	77.8	3.2	78.9	-1.6	80.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	38.3	39.3	43.5	49.8	56.2	61.1	64.5	63.7	59.3	52.4	44.8	39.5	(6)	(7)
	DBStd	11.19	6.82	6.69	6.03	6.53	6.31	5.34	5.30	4.64	4.84	5.73	5.90	7.21	(7)	(8)
	HDD50	1520	364	302	212	84	13	0	0	0	1	41	174	329	(8)	(9)
	HDD65	5222	828	720	666	458	282	142	75	80	181	392	607	791	(9)	(10)
	CDD50	1917	1	2	11	77	205	332	449	425	280	115	17	3	(10)	(11)
	CDD65	143	0	0	0	1	10	24	58	40	10	0	0	0	(11)	(12)
	CDH74	1045	0	0	0	12	83	201	424	264	59	2	0	0	(12)	(13)
	CDH80	239	0	0	0	0	10	48	111	61	9	0	0	0	(13)	(14)
(14) Wind	WSAvg	9.5	11.5	11.1	10.4	9.1	9.1	8.5	8.3	7.7	7.7	9.4	9.9	11.2	(14)	(15)
(15) Precipitation	PrecAvg	34.0	2.8	2.6	2.1	1.8	2.4	2.8	3.5	3.4	3.0	3.0	3.3	3.3	(15)	(16)
	PrecMax	42.4	5.2	5.6	3.8	4.1	5.2	5.8	7.4	8.3	7.9	7.3	5.7	6.4	(16)	(17)
	PrecMin	24.3	0.2	0.6	0.5	0.0	0.9	0.7	0.7	0.7	0.6	0.8	0.4	0.7	(17)	(18)
	PrecStd	4.7	1.4	1.2	1.1	1.0	1.1	1.5	1.7	2.0	1.7	1.4	1.3	1.5	(18)	(19)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.5	56.8	63.9	75.7	81.0	85.9	88.8	86.6	80.1	71.2	61.1	55.8	(19)	(20)
		MCWB	51.9	50.3	54.9	60.7	66.7	70.3	70.7	70.1	68.0	63.3	57.4	53.3	(20)	(21)
	2%	DB	52.0	52.7	59.2	69.8	75.9	79.6	82.8	80.6	75.0	65.6	57.8	53.5	(21)	(22)
		MCWB	49.9	49.3	51.8	58.0	62.2	67.2	68.9	68.4	66.0	60.8	55.3	51.5	(22)	(23)
	5%	DB	49.8	50.6	55.4	65.0	72.1	75.0	78.7	76.3	71.1	62.8	55.5	51.5	(23)	(24)
		MCWB	47.9	47.9	50.6	55.5	60.7	65.1	67.6	66.4	63.8	59.0	53.0	49.5	(24)	(25)
	10%	DB	47.7	48.4	52.5	60.9	67.8	71.4	74.9	72.7	67.7	60.6	53.2	49.3	(25)	(26)
		MCWB	45.9	46.1	48.6	53.7	59.1	63.1	65.8	64.5	61.8	57.4	51.2	47.4	(26)	(27)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.5	52.0	55.8	62.6	68.0	72.2	73.0	73.1	69.8	64.1	58.1	54.1	(27)	(28)
		MCDB	54.1	54.9	61.7	72.8	78.2	82.9	82.8	83.2	77.8	69.4	59.8	55.3	(28)	(29)
	2%	WB	50.2	50.0	53.1	59.0	64.5	68.6	70.6	69.6	66.9	61.5	55.7	51.8	(29)	(30)
		MCDB	51.7	52.1	57.6	67.9	72.3	77.1	80.3	78.1	73.3	65.1	57.7	53.4	(30)	(31)
	5%	WB	48.1	48.1	51.1	56.5	61.8	66.1	68.6	67.5	64.9	59.6	53.5	49.9	(31)	(32)
		MCDB	49.6	50.1	54.7	63.9	69.7	73.3	76.8	74.4	69.7	62.4	55.1	51.4	(32)	(33)
	10%	WB	46.2	46.3	49.1	54.2	59.6	64.0	66.6	65.6	62.9	57.9	51.3	47.7	(33)	(34)
		MCDB	47.7	48.4	52.3	59.9	66.5	70.3	73.1	71.1	66.9	60.2	53.0	49.3	(34)	(35)
(35) Mean Daily Temperature Range	5% DB	MDBR	8.5	10.1	13.1	16.3	16.0	15.9	15.2	15.6	14.6	12.1	9.6	8.1	(35)	(36)
		MCDBR	9.0	12.3	18.0	22.0	22.1	22.2	21.6	21.0	18.8	13.8	10.6	8.4	(36)	(37)
		MCWBR	8.3	9.6	12.2	12.7	11.5	12.0	10.5	10.8	10.6	9.4	8.9	7.7	(37)	(38)
	5% WB	MCDBR	9.0	11.0	16.0	20.5	19.7	19.9	19.5	19.0	16.9	13.3	10.4	8.4	(38)	(39)
		MCWBR	8.5	9.1	11.5	12.4	11.1	11.7	10.3	10.7	10.2	9.6	9.0	7.9	(39)	(40)
(40) Clear-Sky Solar Irradiance	taub		0.344	0.366	0.401	0.426	0.433	0.427	0.437	0.427	0.402	0.380	0.356	0.340	(40)	(41)
	taud		2.293	2.269	2.185	2.160	2.194	2.228	2.210	2.255	2.311	2.332	2.308	2.282	(41)	(42)
	Ebn at Noon		202	233	246	256	260	263	258	253	245	225	194	180	(42)	(43)
	Edh at Noon		21	28	37	43	43	42	42	38	33	26	21	18	(43)	(44)
(44) All-Sky Solar Radiation	RadAvg		226	443	824	1329	1611	1726	1630	1374	1000	577	263	171	(44)	(45)
	RadStd		25	83	104	134	115	123	180	115	88	48	25	20	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page